

TJ Moody

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EDUCATION

UNIVERSITY OF MICHIGAN - College of Engineering

Ann Arbor, MI

B.S.E. - Intended Major: Computer Science; GPA: 3.4

May 2028

- Awarded LEAD Scholarship, Honors
- Coursework: Digital Logic, Discrete Math, Data Structures & Algorithms, Computer Organization, Database Systems, Web Systems, Foundations of Computer Science. In Progress: Computer Security, Operating Systems

SKILLS

- Programming: C/C++, Git/Github, Linux, Sysadmin, macOS, Javascript, Typescript, React, Rust, Lua, Docker, SQL, AI Agents, Bash, x86 Assembly, MATLAB
- Languages: Proficient in Spanish

PERSONAL PROJECT EXPERIENCE

YouTube Clone - github.com/tj-moody/vidtube

Typescript, React, Go, PostgreSQL, Docker, AWS S3, SQS

- Built a full-stack video platform with browser uploads and paginated video browsing through a REST API.
- Implemented direct uploads using presigned URLs, keeping video file transfers off the API server.
- Developed an event-driven Go worker to consume S3 upload notifications through SQS, update video readiness in PostgreSQL, and retry failed database updates for scalability and post-upload asynchronous consumers.
- Containerized the frontend, backend, database, and LocalStack with Docker Compose, automating local S3 bucket, queue, and event notification setup.

Instagram Clone - Class Project

Javascript, React, Python, Flask, HTML, SQL, EC2

- Implemented paginated REST API in Flask supporting CRUD operations for posts, likes, and comments with HTTP Basic Authentication and session-based auth
- Built React frontend with AJAX calls to REST API, rendering a dynamic post feed with real-time like/comment updates and double-click-to-like without page reloads

MapReduce Framework - Class Project

Python, TCP/UDP, Multithreading, Linux, AWS

- Implemented a distributed MapReduce framework in Python with Manager/Worker architecture using TCP/UDP sockets for job scheduling, worker coordination, and heartbeat-based fault detection.
- Engineered parallel map and reduced execution with task partitioning, inter-process communication, and automatic task reassignment upon worker failure.
- Deployed the framework on AWS EC2 instances, enabling distributed execution across multiple virtual machines and validating end-to-end job processing in a cloud environment

RISC-V Assembler

Rust

- Built assembler for RV32I instruction subset with support for labels, directives, and pseudo-instructions
- Validated output correctness against reference implementations and edge cases

Rasterizer - github.com/tj-moody/rasterbox

C++, SFML, GLM

- Read papers on modern rendering algorithms and GPU architecture
- Benchmarked and profiled rasterization methods to optimize performance and learn performant C++ programming

Chess Engine - github.com/tj-moody/tnewt_rs

Rust, perf

- Developed heavily optimized chess move generation
- Verified correctness through robust testing framework against known difficult edge-cases
- Achieved 5x performance improvement through iterative benchmarking

WORK EXPERIENCE

Boston Ultimate Disc Alliance (BUDA)

Boston, MA

Ultimate Frisbee Coach

July-August 2023

- Coached underserved youth in Ultimate Frisbee to facilitate community engagement
- Organized and managed kids ages 8-13, taught them rules, strategies, and basic skills of Ultimate Frisbee

ACTIVITIES

BagnUM Ultimate Frisbee team: Captain

September 2024-Present

Engineering Abroad in Madrid

June-July 2025